Anudeep A

Portfolio|*anudeepa0612@gmail.com* | LinkedIn | **840-888-9831**

PROFESSIONAL SUMMARY

Results-driven Software Engineer with **5+ years of experience designing and building scalable Java microservices and cloud-based applications** across healthcare, banking, and manufacturing domains. Strong expertise in **Java 17, Spring Boot, REST APIs, and event-driven architecture** deployed on **AWS and Azure**. Experienced in building distributed systems using **Docker, Kubernetes, CI/CD pipelines, and modern frontend frameworks including React and Angular**. AWS Certified Solutions Architect Associate and Azure Administrator Associate.

TECHNICAL SKILLS

- **Programming Languages**: Java (8/11/17), JavaScript, TypeScript, SQL, Python, HTML5, CSS3
- **Backend Frameworks & Libraries**: Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Spring Cloud (Eureka, Config Server, API Gateway), Hibernate, RESTful Web Services, Microservices Architecture
- **Frontend Technologies**: Angular, React.js, TypeScript, RxJS, Bootstrap
- **Cloud Platforms**: AWS (EKS, EC2, S3, RDS, Lambda), Azure (AKS, App Service, Azure SQL)
- **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, GitHub Actions, Azure DevOps, Terraform, Helm
- **Databases:** PostgreSQL, MySQL, Oracle Database, MongoDB, Redis, Azure SQL Database
- **Messaging & Event Streaming:** Apache Kafka, RabbitMQ, AWS SNS/SQS, Azure Service Bus
- **Testing Frameworks:** JUnit, Mockito, TestNG, Selenium, Postman
- **Build & Version Control:** Git, GitHub
- **Monitoring & Logging:** Splunk, ELK Stack, CloudWatch, Application Insights, Prometheus, Grafana
- **Tools & Methodologies:** JIRA, Confluence, Swagger/OpenAPI, SonarQube, IntelliJ IDEA, Eclipse, Visual Studio Code, Agile/Scrum, Kanban

PROFESSIONAL EXPERIENCE

Insmed Incorporated

Software Engineer | US | JUN 2025 – PRESENT

- Led migration of **15+ legacy healthcare applications** from on-prem infrastructure to **AWS EKS**, modernizing clinical trial and patient management platforms and reducing infrastructure costs by **$2.1M annually (35%)** while improving system availability to **99.95% uptime**.
- Architected and developed **12 Java 17 Spring Boot microservices** supporting patient data workflows and regulatory compliance modules, implementing service discovery and API gateway patterns to improve system scalability and maintainability.
- Designed **event-driven microservices architecture using Apache Kafka**, processing **500K+ healthcare events daily** (patient updates, lab results, drug interactions) while ensuring reliable message delivery and idempotent event processing.
- Implemented **OAuth2 and JWT authentication using Spring Security** to enforce role-based access control and protect sensitive PHI data, ensuring system compliance with **HIPAA security standards**.
- Built **CI/CD pipelines using Jenkins and GitHub Actions**, automating builds, security scans, and container deployments with Docker and Kubernetes, reducing release cycles from **2 weeks to 3 days**.
- Optimized **PostgreSQL database performance** through indexing, query tuning, and partitioning strategies, reducing API latency by **40%** for high-traffic patient dashboards used by **50K+ daily users**.
- Implemented **Redis distributed caching** for frequently accessed clinical reference data, reducing database load by **60%** and significantly improving system throughput during peak research activity.
- Developed **React-based clinical trial management dashboard** with reusable components and REST API integrations, enabling researchers and healthcare administrators to track patient enrollment and trial progress in real time.
- Automated AWS infrastructure provisioning using **Terraform**, managing EKS clusters, RDS databases, IAM roles, and S3 storage to maintain consistent environments across development, staging, and production.
- Implemented **Kubernetes Helm deployments with horizontal auto-scaling**, enabling the platform to handle large spikes in clinical trial enrollment workloads without service interruptions.

- Established centralized observability using **ELK Stack and AWS CloudWatch**, building dashboards for application health monitoring, log analysis, and proactive incident detection.
- Mentored **3 junior developers** on microservices architecture and Spring Boot best practices, improving overall code quality and increasing **SonarQube quality gate scores by 25%**

Environment: Java 17, Spring Boot 3.2, Spring Cloud, Spring Security, Spring Data JPA, Microservices, Apache Kafka, OAuth2/JWT, AWS (EKS, EC2, S3, RDS PostgreSQL, CloudWatch, IAM), Docker, Kubernetes, Helm, Terraform, Jenkins, GitHub Actions, Redis, React, PostgreSQL, JUnit, Mockito, SonarQube, ELK Stack, JIRA, Agile/Scrum

Bank of New York Mellon
Java Full Stack Developer | US | JUN 2024 – JUN 2025

- Developed and maintained **8 Spring Boot microservices** supporting global banking operations including trade settlement, wire transfers, and ACH processing, handling **$2.3M+ transactions daily** with **99.97% platform uptime**.
- Migrated **5 legacy J2EE applications to microservices architecture** using Spring Boot and Spring Cloud, enabling independent deployments and reducing **system recovery time from 4 hours to 30 minutes**.
- Designed and implemented **40+ REST APIs** following OpenAPI standards to support internal banking systems and external integrations with payment networks and financial clearing services.
- Implemented **event-driven architecture using Apache Kafka**, enabling asynchronous communication across payment processing, fraud detection, and compliance services while processing **1.2M+ events daily**.
- Developed **Spring Batch pipelines** for end-of-day financial reconciliation, processing **500K+ transaction records** and generating automated settlement reports for operations teams.
- Implemented **OAuth2 / OpenID Connect authentication with Spring Security and JWT**, integrating with enterprise Active Directory for secure API access and single sign-on capabilities.
- Built **CI/CD pipelines using Azure DevOps**, automating builds, testing, Docker image creation, and deployments to **Azure Kubernetes Service (AKS)**, reducing deployment time by **60%**.
- Optimized **Oracle and Azure SQL database performance** through query tuning, indexing, and connection pooling with HikariCP, improving API response times by **35%** for high-frequency trading services.
- Developed **React-based trading and operations dashboards**, enabling real-time transaction monitoring, advanced search, and interactive reporting for compliance and operations teams.
- Implemented **WebSocket-based real-time updates** for trade status notifications and payment confirmations, improving operational visibility without requiring page refreshes.
- Strengthened platform security by implementing **PCI-DSS and SOX compliance controls**, including data encryption, audit logging, and role-based data masking for sensitive financial information.
- Built monitoring and distributed tracing dashboards using **Azure Application Insights and Splunk**, improving incident detection and reducing production troubleshooting time.
- Participated in **24/7 production support rotation**, resolving high-priority incidents within SLA and achieving **98.5% first-contact resolution rate**.

Environment: Java 11, Spring Boot 2.7, Spring Cloud, Spring Security, Spring Data JPA, Microservices, Apache Kafka, REST APIs, OAuth2/JWT, Azure (AKS, App Service, Azure SQL, Key Vault, Application Insights), Azure DevOps, Docker, Kubernetes, Oracle 19c, HikariCP, React, TypeScript, WebSocket, Maven, JUnit, Mockito, TestNG, Swagger/OpenAPI, Splunk, JIRA, Agile/Scrum

Regal Rexnord Corporation
Backend Java Developer | US | JAN 2024 – JUN 2024

- Developed **Java 11 Spring Boot microservices** for an Industrial IoT platform monitoring **25K+ connected machines across 500+ manufacturing facilities**, enabling real-time telemetry processing and predictive maintenance analytics.
- Designed **REST APIs using Spring Boot and Spring Data JPA** to ingest and process **100K+ sensor readings per minute** from PLC and SCADA systems, supporting device registration, equipment diagnostics, and maintenance scheduling workflows.
- Implemented **AWS IoT Core and Lambda-based event pipelines** to process streaming telemetry data and detect anomaly patterns, enabling early equipment failure detection and reducing unplanned downtime by **30%**.
- Built **Angular-based operational dashboards** for plant engineers and technicians, enabling real-time monitoring of machine health, predictive alerts, and maintenance insights through REST API integrations.
- Designed **PostgreSQL time-series schemas using TimescaleDB** with partitioning and indexing strategies, improving analytics query performance by **25%** for equipment health reports and operational dashboards.

- Implemented **Redis caching layer** for equipment metadata and configuration data, improving API response times by **50%** for technician mobile applications.
- Containerized services using **Docker** and deployed them on **AWS ECS Fargate**, enabling scalable microservice deployments across multiple production environments.
- Configured **AWS Auto Scaling and Application Load Balancer** to dynamically scale container workloads during peak production hours, automatically increasing service capacity from **4 to 20 instances**.
- Built **CI/CD pipelines using Jenkins and AWS CodePipeline**, automating unit testing, container builds, and blue-green deployments, improving deployment reliability and reducing release risks.
- Automated cloud infrastructure provisioning using **AWS CloudFormation**, managing VPC networks, ECS clusters, RDS databases, and IAM policies for secure multi-environment deployments.
- Implemented centralized logging and monitoring using **CloudWatch Logs and metrics dashboards**, enabling proactive incident detection and faster troubleshooting for production systems.
- Produced **Swagger/OpenAPI documentation** for device telemetry and maintenance APIs, improving integration speed for downstream analytics services and mobile applications.

Environment: Java 11, Spring Boot 2.6, Spring Data JPA, Microservices, REST APIs, AWS (IoT Core, Lambda, ECS Fargate, S3, RDS PostgreSQL, CloudFormation, Auto Scaling, ALB, CloudWatch, SNS, IAM), Docker, PostgreSQL 13, TimescaleDB, Redis, Jenkins, AWS CodePipeline, Maven, JUnit, Mockito, Swagger/OpenAPI, Angular, Git, JIRA, Agile/Scrum

Cognizant Technology Solutions
Software Engineer | India | AUG 2021 – MAY 2022

- Developed and enhanced **customer-facing online banking platform** used by **2M+ retail banking customers**, supporting account management, bill payments, fund transfers, and loan services for a tier-1 US banking client.
- Built backend services using **Java 8, Spring MVC, and Hibernate**, implementing modules for customer profiles, multi-account aggregation, transaction history retrieval, and statement generation.
- Designed and implemented **RESTful APIs using JAX-RS (Jersey)** to enable seamless integration with mobile banking applications and external financial service providers.
- Integrated banking systems with **payment networks and vendor platforms** (Visa Direct, Mastercard Send, FIS/Fiserv) to support real-time bill payments and peer-to-peer fund transfers.
- Developed **Angular-based UI components** for account dashboards, transaction search, and bill payment workflows, improving user interaction and reducing page reloads through asynchronous API calls.
- Implemented **scheduled batch processes using Quartz and Spring Scheduler** for automated tasks including recurring payments, statement generation, and notification services.
- Optimized **MySQL database performance** through indexing and query tuning, improving page load times by **20%** for account summary and transaction history features.
- Enforced **secure transaction handling using Spring transaction management**, ensuring ACID compliance for critical financial operations such as transfers and bill payments.
- Strengthened application security by implementing **input validation and OWASP best practices**, protecting the platform from SQL injection, XSS, and CSRF vulnerabilities.
- Developed **unit and integration tests using JUnit and Mockito**, achieving **85%+ code coverage** and improving system stability during continuous integration builds.
- Integrated applications build with **Jenkins CI pipelines**, automating compilation, testing, and deployment of WAR artifacts to WebLogic servers in development and QA environments.
- Collaborated with cross-functional teams in **Agile Scrum environment**, supporting production releases, resolving defects, and contributing to feature delivery across multiple sprint cycles.

Environment: Java 8, Spring MVC, Hibernate, JAX-RS (Jersey), REST APIs, Quartz Scheduler, MySQL, Angular, HTML5, CSS3, JavaScript, Maven, JUnit, Mockito, Jenkins, Git, WebLogic, JIRA, Agile/Scrum

Birlasoft India
Java Developer | India | JAN 2021 – JUL 2021

- Developed **Java-based enterprise applications** for telecom CRM and retail e-commerce platforms, implementing core business services for customer onboarding, plan management, billing inquiries, and support ticket workflows.
- Designed and implemented **Spring Boot REST APIs** handling **50K+ API requests daily**, supporting telecom customer onboarding, plan subscriptions, and billing operations.

- Built **ETL pipelines using Spring Batch** to process customer data from legacy systems, transforming and loading records into PostgreSQL data warehouse for reporting and analytics.
- Developed batch processing jobs for **monthly billing and invoice generation**, processing **200K+ customer accounts** and automating charge calculations and statement creation.
- Designed relational and document database models using **PostgreSQL and MongoDB**, supporting transactional telecom operations and scalable product catalog services.
- Implemented **JWT-based authentication and RBAC using Spring Security**, securing REST APIs and enabling secure integration with partner applications.
- Developed **Angular-based UI components** for CRM dashboards including customer onboarding forms, plan management interfaces, and dynamic reporting tables.
- Containerized applications using **Docker** for local development and testing environments, improving environment consistency across developer workstations.
- Implemented centralized logging using **SLF4J and Log4j2**, enabling structured log tracing and improving debugging for distributed application components.
- Collaborated with DevOps teams to deploy applications on **Linux-based servers using Apache Tomcat**, supporting internal enterprise systems and staging environments.
- Contributed to code quality initiatives using **SonarQube and peer code reviews**, improving code coverage and reducing production defects across releases.

Environment: Java 8/11, Spring Boot 2.4, Spring MVC, Spring Data JPA, Spring Batch, Spring Security, REST APIs, JWT, PostgreSQL, MongoDB, Angular 11, Docker, Maven, JUnit, Mockito, Git, SonarQube, SLF4J, Log4j2, Apache Tomcat, Linux (RHEL), JIRA, Agile/Scrum

Sonata Software Solutions
Software Development Intern | India | MAY 2020 – DEC 2020

- Contributed to development of **Java-based travel booking platform** supporting flight and hotel reservations for an online travel agency serving travelers across the Asia-Pacific region.
- Developed backend components using **Java 8 and Spring MVC**, implementing business logic for flight search, itinerary management, fare calculation, and booking confirmation workflows.
- Built **REST APIs** to support web and mobile applications, enabling integration with external airline distribution systems for real-time flight availability and pricing validation.
- Designed database tables and SQL queries in **MySQL** for booking data, passenger details, and payment records, ensuring reliable transaction handling for reservation workflows.
- Developed responsive UI pages using **HTML, CSS, JavaScript, and Bootstrap**, implementing dynamic search results and booking flows for the web portal.
- Wrote **JUnit tests for service-layer components** and collaborated with senior developers in code reviews to improve code quality and follow enterprise development practices.

Environments: Java 8, Spring MVC, REST APIs, MySQL, HTML, CSS, JavaScript, Bootstrap, Maven, JUnit, Apache Tomcat, Git, JIRA, Agile/Scrum

EDUCATION

- Master's in computer sciences |**University of Central Missouri**| Warrensburg, MO| AUG 2022 to MAY 2024
- Bachelor of Technology in Engineering | **VR Siddhartha Engineering** | India | MAY 2017 to MAY 2021

CERTIFICATIONS:

- AWS Certified Solutions Architect – Associate | Amazon Web Services
- Microsoft Certified: Azure Administrator Associate | Microsoft Azure